Airline Customer Satisfaction Prediction Using Logistic Regression

Project Overview

This project develops a Logistic Regression model to predict airline customer satisfaction using passenger survey data. The objective is to identify the factors that most strongly influence satisfaction and provide actionable business recommendations.

1. Import Libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    confusion_matrix,
    classification_report,
    ConfusionMatrixDisplay,
    roc_curve,
    roc_auc_score
)

plt.style.use('default')
pd.set_option('display.max_columns', None)
```

2. Load Dataset

In [2]:
```
df = pd.read_csv('Invistico_Airline.csv')

print("Dataset Shape:", df.shape)
df.head()
```

```
Dataset Shape: (129880, 22)
```

Out[2]:

	satisfaction	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arrival time convenient	F d
0	satisfied	Loyal Customer	65	Personal Travel	Eco	265	0	0	
1	satisfied	Loyal Customer	47	Personal Travel	Business	2464	0	0	
2	satisfied	Loyal Customer	15	Personal Travel	Eco	2138	0	0	
3	satisfied	Loyal Customer	60	Personal Travel	Eco	623	0	0	
4	satisfied	Loyal Customer	70	Personal Travel	Eco	354	0	0	

3. Data Understanding

In [3]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 129880 entries, 0 to 129879
Data columns (total 22 columns):
 #   Column                             Non-Null Count   Dtype  
---  ------                             --------------   -----  
 0   satisfaction                       129880 non-null  object 
 1   Customer Type                      129880 non-null  object 
 2   Age                                129880 non-null  int64  
 3   Type of Travel                     129880 non-null  object 
 4   Class                              129880 non-null  object 
 5   Flight Distance                    129880 non-null  int64  
 6   Seat comfort                       129880 non-null  int64  
 7   Departure/Arrival time convenient  129880 non-null  int64  
 8   Food and drink                     129880 non-null  int64  
 9   Gate location                      129880 non-null  int64  
 10  Inflight wifi service              129880 non-null  int64  
 11  Inflight entertainment             129880 non-null  int64  
 12  Online support                     129880 non-null  int64  
 13  Ease of Online booking             129880 non-null  int64  
 14  On-board service                   129880 non-null  int64  
 15  Leg room service                   129880 non-null  int64  
 16  Baggage handling                   129880 non-null  int64  
 17  Checkin service                    129880 non-null  int64  
 18  Cleanliness                        129880 non-null  int64  
 19  Online boarding                    129880 non-null  int64  
 20  Departure Delay in Minutes         129880 non-null  int64  
 21  Arrival Delay in Minutes           129487 non-null  float64
dtypes: float64(1), int64(17), object(4)
memory usage: 21.8+ MB
```

In [4]:
```
df.describe().T
```

Out[4]:

	count	mean	std	min	25%	50%	75%	max
Age	129880.0	39.427957	15.119360	7.0	27.0	40.0	51.0	85.0
Flight Distance	129880.0	1981.409055	1027.115606	50.0	1359.0	1925.0	2544.0	6951.0
Seat comfort	129880.0	2.838597	1.392983	0.0	2.0	3.0	4.0	5.0
Departure/Arrival time convenient	129880.0	2.990645	1.527224	0.0	2.0	3.0	4.0	5.0
Food and drink	129880.0	2.851994	1.443729	0.0	2.0	3.0	4.0	5.0
Gate location	129880.0	2.990422	1.305970	0.0	2.0	3.0	4.0	5.0
Inflight wifi service	129880.0	3.249130	1.318818	0.0	2.0	3.0	4.0	5.0
Inflight entertainment	129880.0	3.383477	1.346059	0.0	2.0	4.0	4.0	5.0
Online support	129880.0	3.519703	1.306511	0.0	3.0	4.0	5.0	5.0
Ease of Online booking	129880.0	3.472105	1.305560	0.0	2.0	4.0	5.0	5.0
On-board service	129880.0	3.465075	1.270836	0.0	3.0	4.0	4.0	5.0
Leg room service	129880.0	3.485902	1.292226	0.0	2.0	4.0	5.0	5.0
Baggage handling	129880.0	3.695673	1.156483	1.0	3.0	4.0	5.0	5.0
Checkin service	129880.0	3.340807	1.260582	0.0	3.0	3.0	4.0	5.0
Cleanliness	129880.0	3.705759	1.151774	0.0	3.0	4.0	5.0	5.0
Online boarding	129880.0	3.352587	1.298715	0.0	2.0	4.0	4.0	5.0
Departure Delay in Minutes	129880.0	14.713713	38.071126	0.0	0.0	0.0	12.0	1592.0
Arrival Delay in Minutes	129487.0	15.091129	38.465650	0.0	0.0	0.0	13.0	1584.0

In [5]:
```
df.isnull().sum()
```

Out[5]:
```
satisfaction                          0
Customer Type                         0
Age                                   0
Type of Travel                        0
Class                                 0
Flight Distance                       0
Seat comfort                          0
Departure/Arrival time convenient     0
Food and drink                        0
Gate location                         0
Inflight wifi service                 0
Inflight entertainment                0
Online support                        0
Ease of Online booking                0
On-board service                      0
Leg room service                      0
Baggage handling                      0
Checkin service                       0
Cleanliness                           0
Online boarding                       0
Departure Delay in Minutes            0
Arrival Delay in Minutes            393
dtype: int64
```

Missing Value Treatment

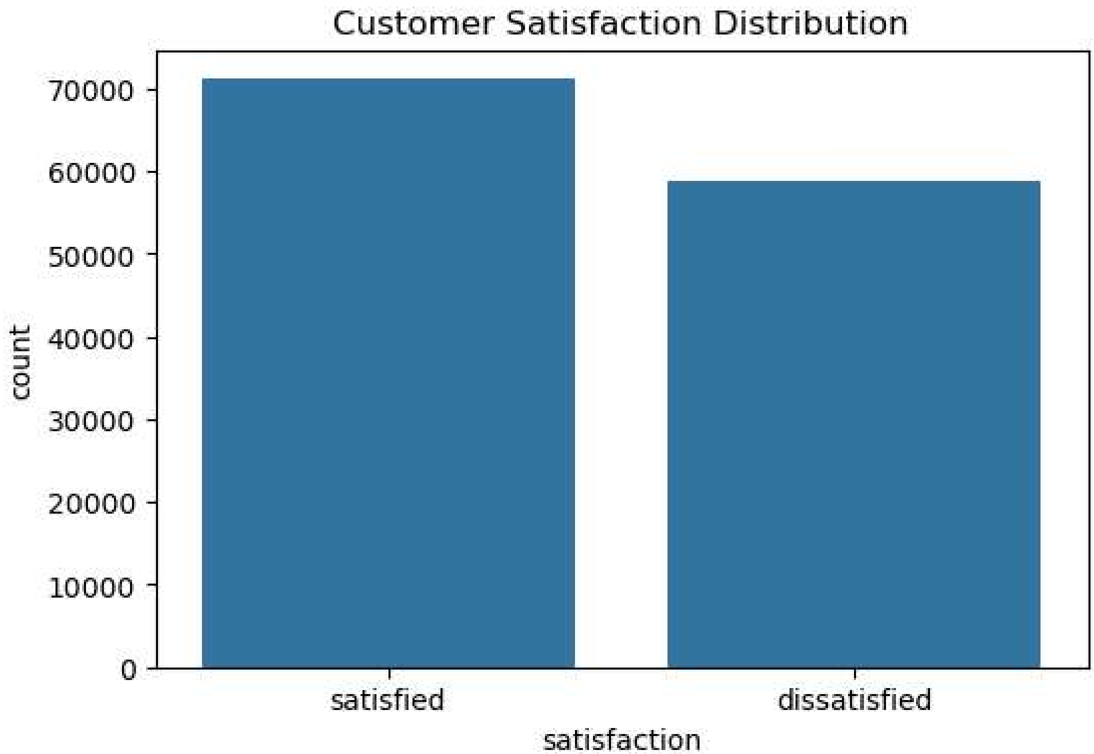
The only missing values occur in 'Arrival Delay in Minutes'. We replace them with the median because delay distributions are usually skewed.

In [6]:
```
df['Arrival Delay in Minutes'] = df['Arrival Delay in Minutes'].fillna(
    df['Arrival Delay in Minutes'].median()
)

df.isnull().sum().sum()
```

Out[6]:
```
np.int64(0)
```

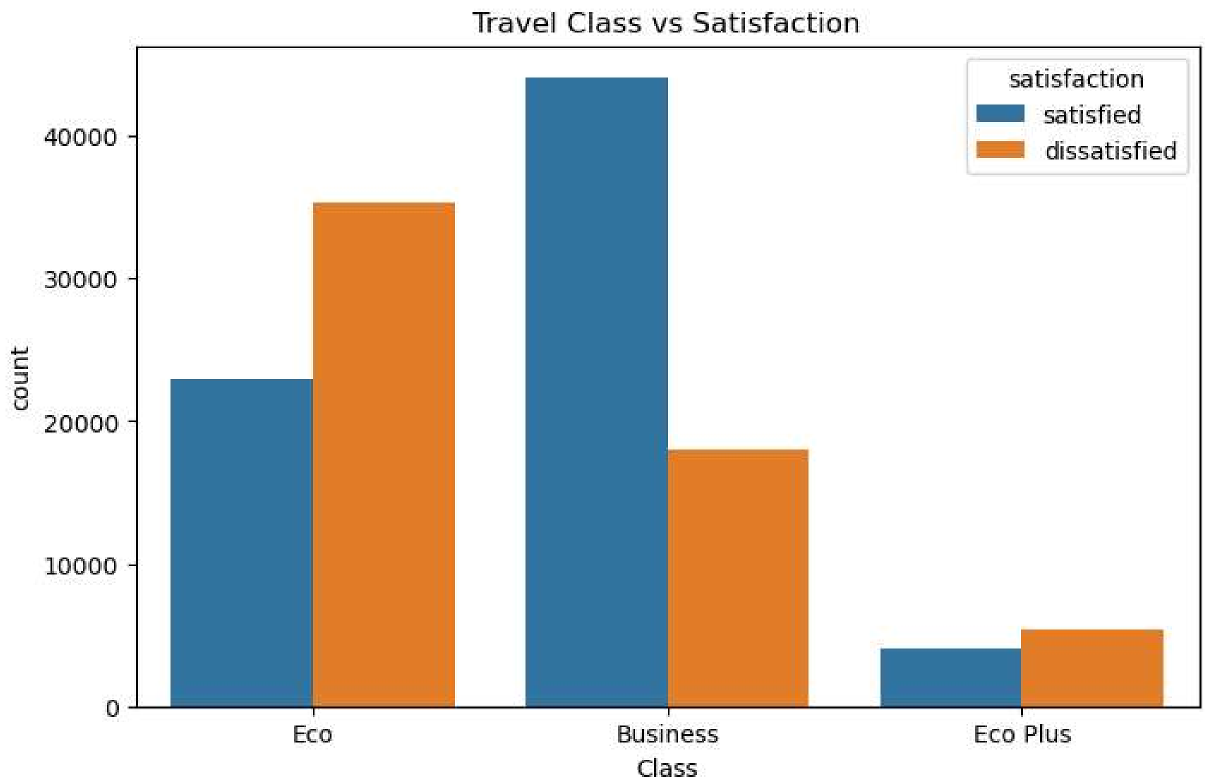
4. Exploratory Data Analysis

In [7]:
```
plt.figure(figsize=(6,4))
sns.countplot(x='satisfaction', data=df)
plt.title('Customer Satisfaction Distribution')
plt.show()

df['satisfaction'].value_counts()
```

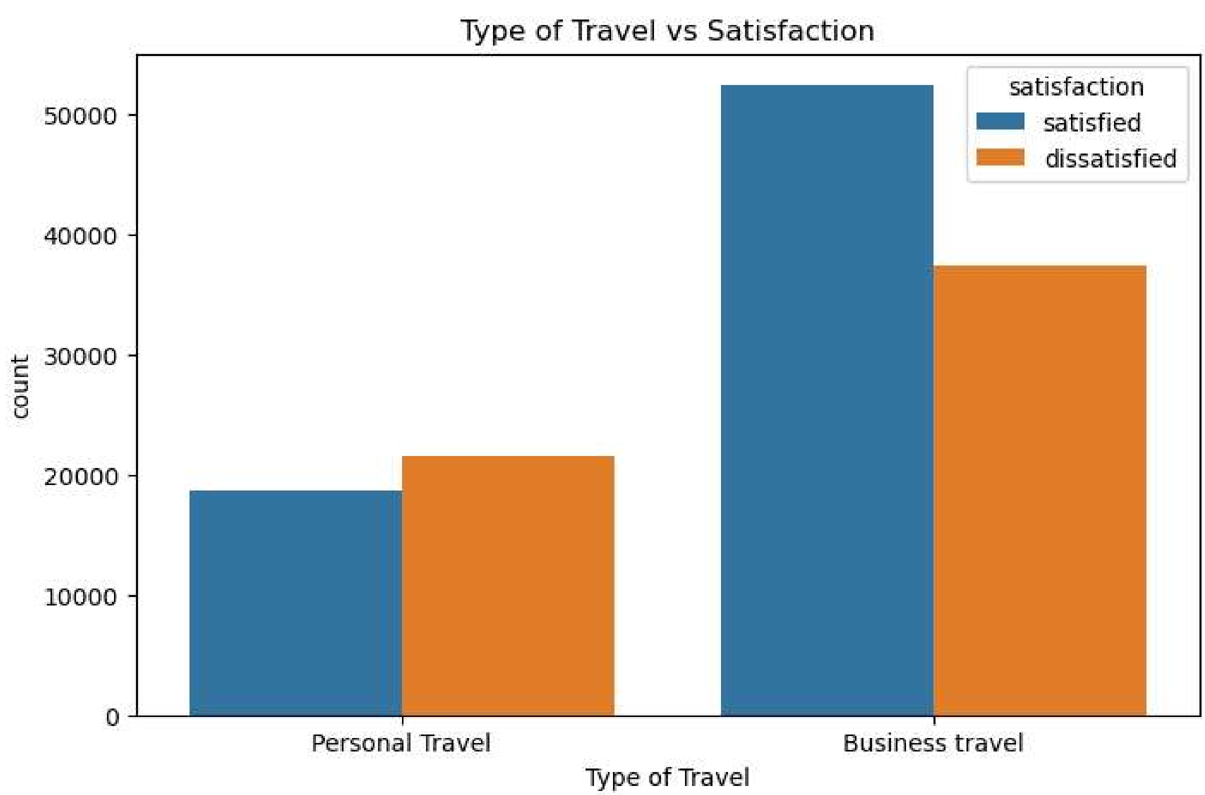


Out[7]:
```
satisfaction
satisfied       71087
dissatisfied    58793
Name: count, dtype: int64
```

In [8]:
```
plt.figure(figsize=(8,5))
sns.countplot(x='Class', hue='satisfaction', data=df)
plt.title('Travel Class vs Satisfaction')
plt.show()
```

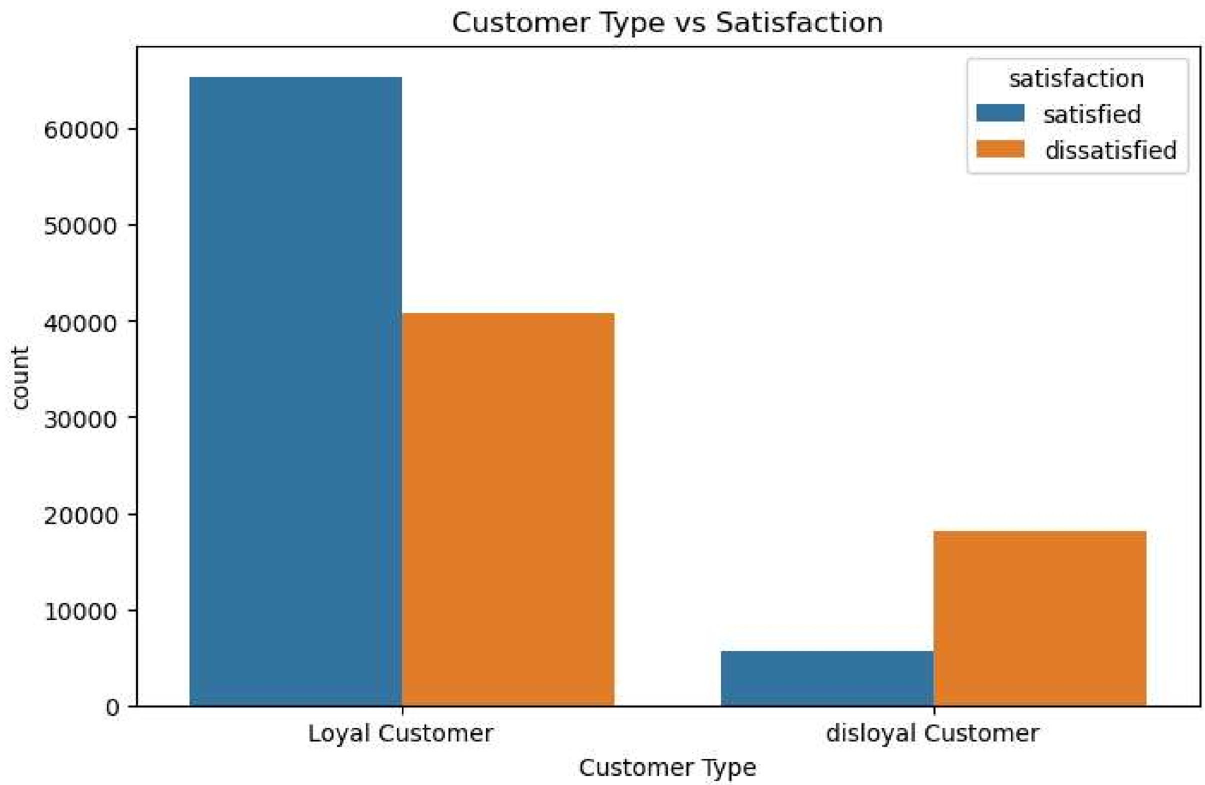


In [9]:
```
plt.figure(figsize=(8,5))
sns.countplot(x='Type of Travel', hue='satisfaction', data=df)
plt.title('Type of Travel vs Satisfaction')
plt.show()
```



In [10]:
```
plt.figure(figsize=(8,5))
sns.countplot(x='Customer Type', hue='satisfaction', data=df)
```

```
plt.title('Customer Type vs Satisfaction')
plt.show()
```



5. Data Preprocessing

In [11]:
```
df['satisfaction'] = df['satisfaction'].map({
    'dissatisfied':0,
    'satisfied':1
})

categorical_cols = ['Customer Type','Type of Travel','Class']

encoders = {}

for col in categorical_cols:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])
    encoders[col] = le
```

6. Feature Selection

In [12]:
```
X = df.drop('satisfaction', axis=1)
y = df['satisfaction']

print(X.shape)
```

```
(129880, 21)
```

7. Train-Test Split

In [13]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print(X_train.shape)
print(X_test.shape)
```

```
(103904, 21)
(25976, 21)
```

Feature Scaling

Logistic Regression performs better when numerical features are standardized.

In [14]:
```
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

8. Build Logistic Regression Model

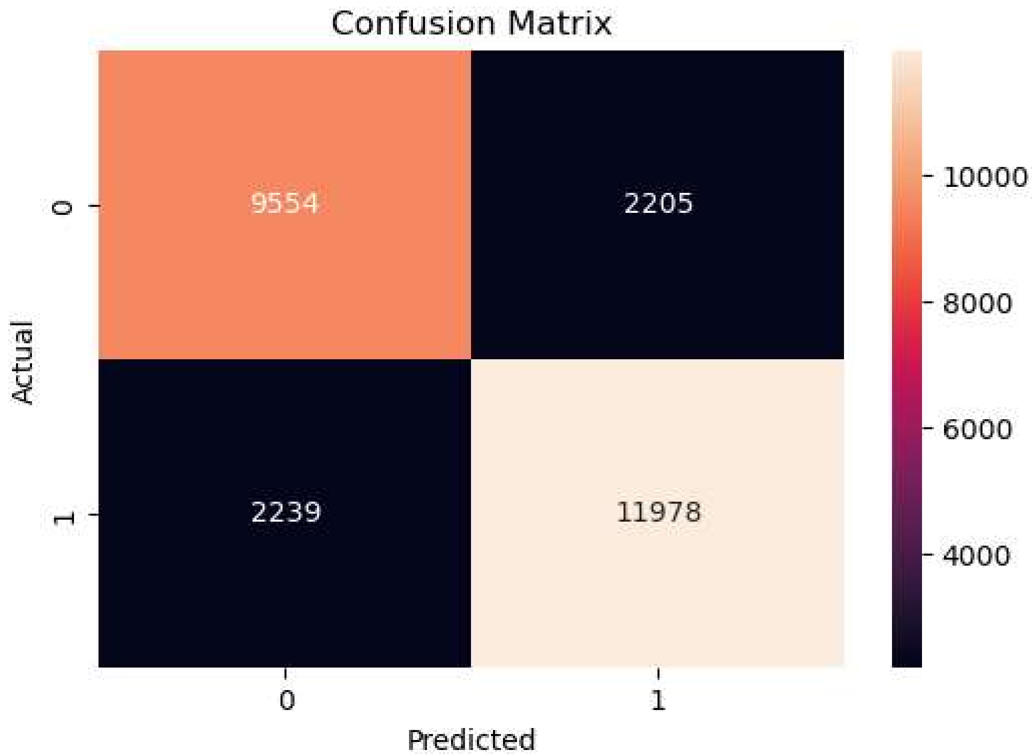
In [15]:
```
model = LogisticRegression(max_iter=2000)

model.fit(X_train_scaled, y_train)

y_pred = model.predict(X_test_scaled)
y_prob = model.predict_proba(X_test_scaled)[:,1]
```

9. Model Evaluation

In [16]:
```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)

print(f'Accuracy : {accuracy:.4f}')
print(f'Precision: {precision:.4f}')
print(f'Recall   : {recall:.4f}')
```

```
Accuracy : 0.8289
Precision: 0.8445
Recall   : 0.8425
```

In [17]:
```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d')
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```



In [18]:
```
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.81      0.81      0.81     11759
           1       0.84      0.84      0.84     14217

    accuracy                           0.83     25976
   macro avg       0.83      0.83      0.83     25976
weighted avg       0.83      0.83      0.83     25976
```

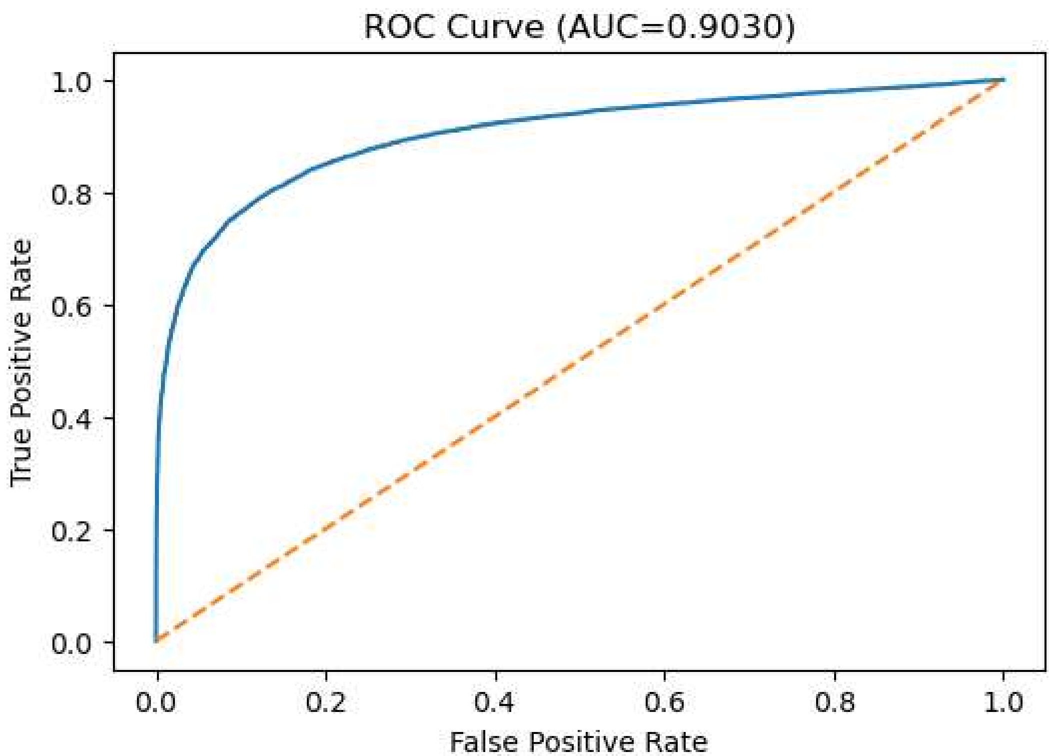
Precision Interpretation

Precision measures the percentage of passengers predicted as satisfied who were actually satisfied.

Recall Interpretation

Recall measures the percentage of truly satisfied passengers correctly identified by the model.

In [19]:
```
auc = roc_auc_score(y_test, y_prob)

fpr, tpr, thresholds = roc_curve(y_test, y_prob)

plt.figure(figsize=(6,4))
plt.plot(fpr, tpr)
plt.plot([0,1],[0,1],'--')
plt.title(f'ROC Curve (AUC={auc:.4f})')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.show()

print("AUC:", auc)
```




```
AUC: 0.9029688965160622
```

10. Coefficient Interpretation

In [20]:
```
coefficients = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_[0]
})

coefficients = coefficients.sort_values(
    by='Coefficient',
    ascending=False
)
```

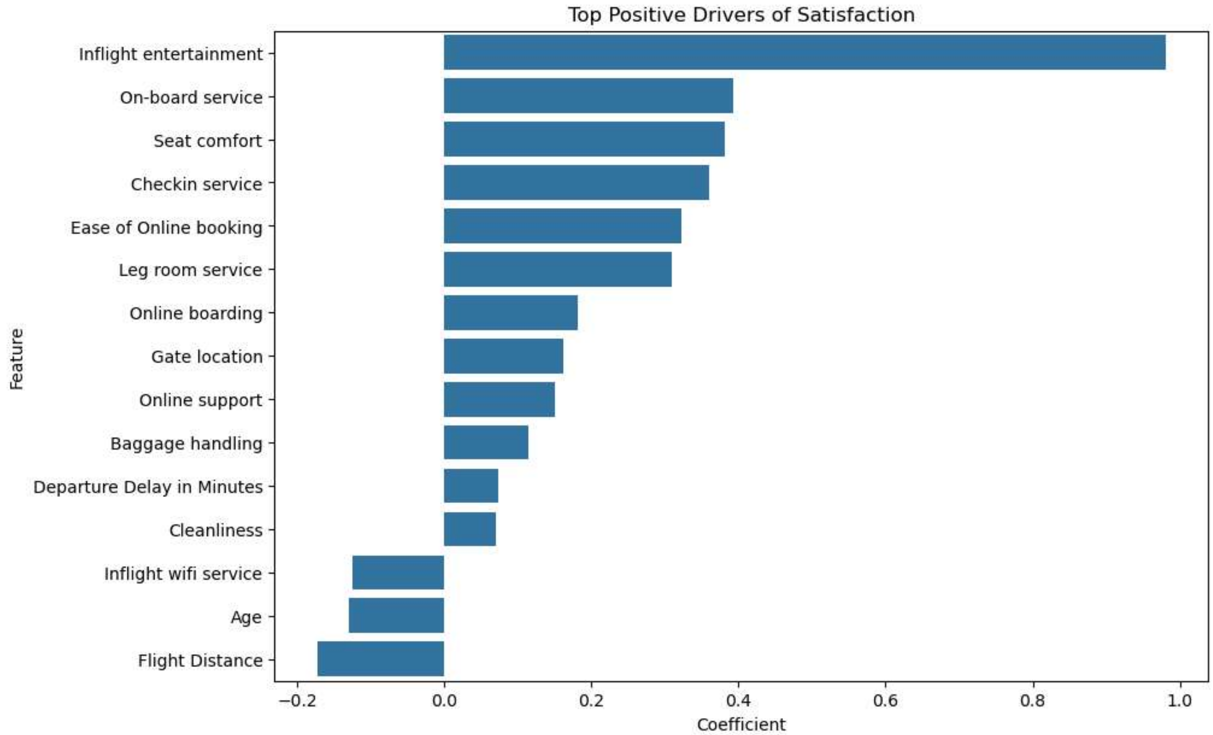
```
coefficients.head(15)
```

Out[20]:

	Feature	Coefficient
10	Inflight entertainment	0.980771
13	On-board service	0.393420
5	Seat comfort	0.382090
16	Checkin service	0.359930
12	Ease of Online booking	0.323253
14	Leg room service	0.310017
18	Online boarding	0.181789
8	Gate location	0.161373
11	Online support	0.151010
15	Baggage handling	0.114572
19	Departure Delay in Minutes	0.074304
17	Cleanliness	0.071090
9	Inflight wifi service	-0.125191
1	Age	-0.129818
4	Flight Distance	-0.172818

In [21]:

```
plt.figure(figsize=(10,7))
sns.barplot(
    data=coefficients.head(15),
    x='Coefficient',
    y='Feature'
)
plt.title('Top Positive Drivers of Satisfaction')
plt.show()
```



In [22]:
```
coefficients.tail(15)
```

Out[22]:

	Feature	Coefficient
18	Online boarding	0.181789
8	Gate location	0.161373
11	Online support	0.151010
15	Baggage handling	0.114572
19	Departure Delay in Minutes	0.074304
17	Cleanliness	0.071090
9	Inflight wifi service	-0.125191
1	Age	-0.129818
4	Flight Distance	-0.172818
20	Arrival Delay in Minutes	-0.265061
7	Food and drink	-0.294136
3	Class	-0.294970
6	Departure/Arrival time convenient	-0.325437
2	Type of Travel	-0.414641
0	Customer Type	-0.751735

Odds Ratio Interpretation

In [23]:
```
odds_ratios = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_[0]
})

odds_ratios["Odds Ratio"] = np.exp(
    odds_ratios["Coefficient"]
)

odds_ratios.sort_values(
    by="Odds Ratio",
    ascending=False
).head(15)
```

Out[23]:

	Feature	Coefficient	Odds Ratio
10	Inflight entertainment	0.980771	2.666512
13	On-board service	0.393420	1.482040
5	Seat comfort	0.382090	1.465345
16	Checkin service	0.359930	1.433229
12	Ease of Online booking	0.323253	1.381615
14	Leg room service	0.310017	1.363448
18	Online boarding	0.181789	1.199361
8	Gate location	0.161373	1.175124
11	Online support	0.151010	1.163008
15	Baggage handling	0.114572	1.121394
19	Departure Delay in Minutes	0.074304	1.077134
17	Cleanliness	0.071090	1.073678
9	Inflight wifi service	-0.125191	0.882329
1	Age	-0.129818	0.878256
4	Flight Distance	-0.172818	0.841291

Odds Ratio Interpretation

Odds ratios provide a more intuitive interpretation of Logistic Regression coefficients.

- Odds Ratio > 1 indicates increased likelihood of satisfaction.
- Odds Ratio < 1 indicates decreased likelihood of satisfaction.

For example, if a feature has an odds ratio of 1.5, a one-unit increase in that feature increases the odds of customer satisfaction by approximately 50%.

11. Business Recommendations

Recommendation 1

Improve inflight Wi-Fi quality and reliability.

Recommendation 2

Enhance online support and booking systems.

Recommendation 3

Improve seat comfort and onboard experience.

Recommendation 4

Reduce departure and arrival delays.

Recommendation 5

Focus on customer experience initiatives for frequent and business travelers.

12. Limitations

1. Logistic Regression assumes a linear relationship in the log-odds.
2. Survey responses may contain subjective bias.
3. Correlated service-rating variables may affect coefficient interpretation.
4. Future customer preferences may differ from historical data.

13. Future Work

1. Random Forest Classifier
2. Gradient Boosting
3. XGBoost
4. Hyperparameter Tuning
5. Customer Segmentation
6. Real-Time Prediction Dashboard

14 Feature Engineering Discussion

Feature Engineering

Several preprocessing steps were performed before model development:

1. Missing values in Arrival Delay in Minutes were replaced using the median value.
2. Categorical variables were encoded using Label Encoding.
3. The target variable (satisfaction) was converted into binary form:
   - Satisfied = 1
   - Dissatisfied = 0
4. Features were standardized using StandardScaler to improve Logistic Regression performance.
5. The dataset was split into training and testing sets using stratified sampling to preserve class distribution.

15. Summary

Executive Summary

This project developed a Logistic Regression model to predict airline customer satisfaction using passenger survey data. After preprocessing, encoding categorical variables, and splitting the data into training and testing sets, a Logistic Regression classifier was trained and evaluated using Accuracy, Precision, Recall, Confusion Matrix, and ROC-AUC metrics.

The analysis identified service-related factors such as Inflight Wi-Fi Service, Online Boarding, Online Support, Seat Comfort, and Inflight Entertainment as major drivers of customer satisfaction. The results provide actionable insights for airline management to improve customer experience and increase passenger retention.